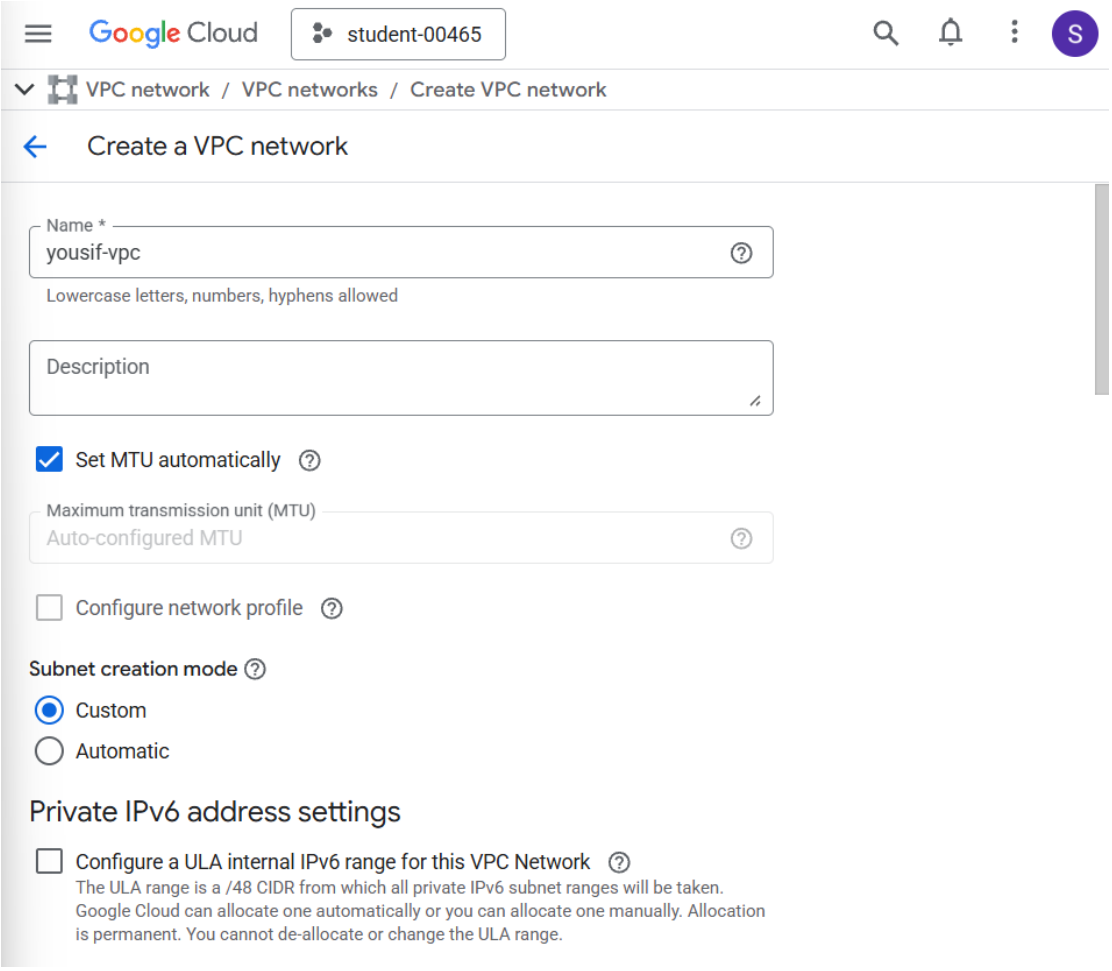
Create VPC Network:



Google Cloud student-00465

Navigation menu (.) rk / VPC networks / Create VPC network

Create a VPC network

New subnet

Name *
public-subnet
Lowercase letters, numbers, hyphens allowed

Description

Region *
us-central1 (Iowa)

IP stack type

- (●) IPv4 (single-stack)
- ( ) IPv4 and IPv6 (dual-stack)
- ( ) IPv6 (single-stack)

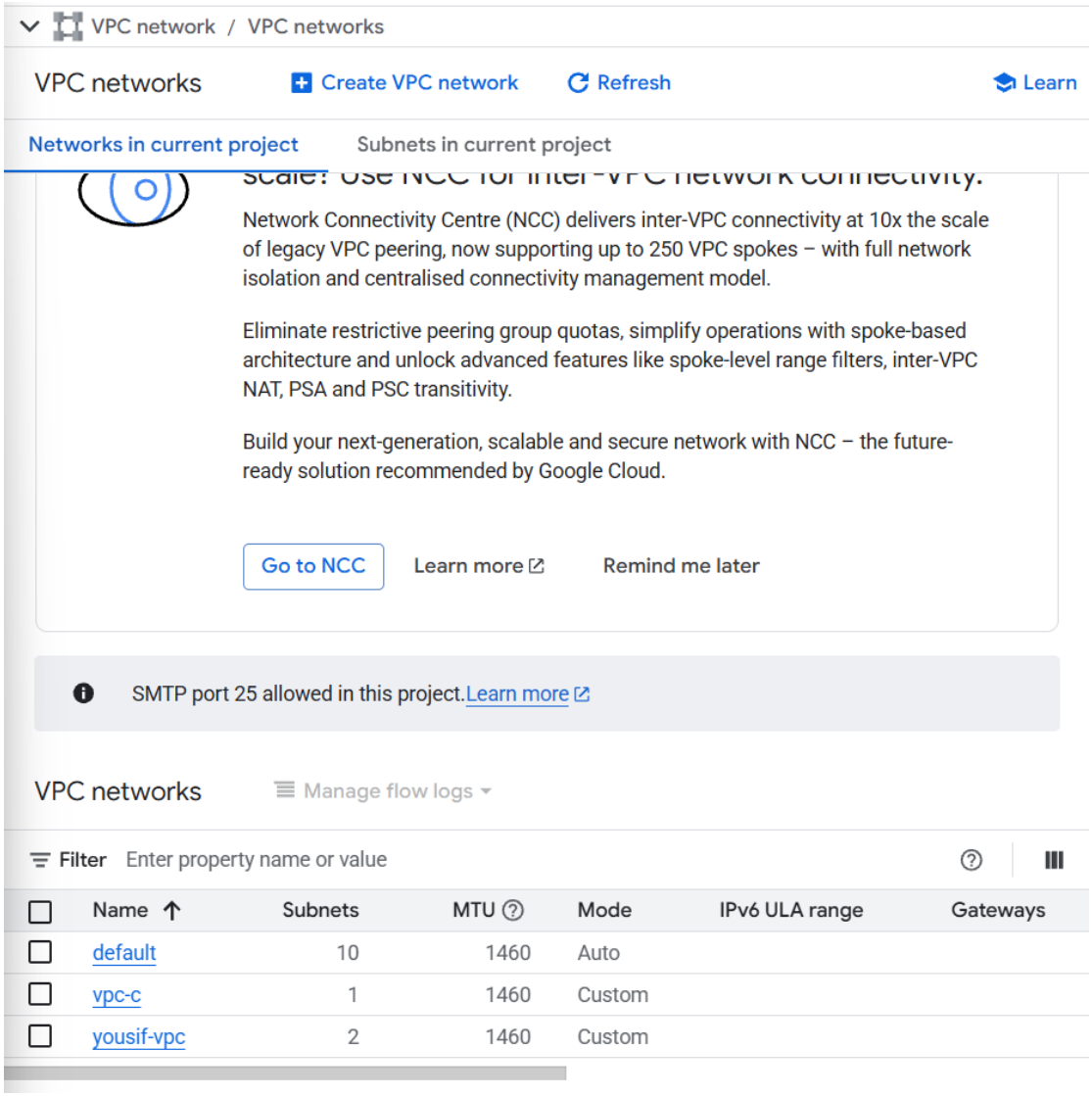
Primary IPv4 range

[ ] Associate with an internal range
Use an internal range to specify the subnet's internal IP address range. The subnetwork can be associated with an entire internal range or only part of the range.

IPv4 range *
10.0.1.0/24
E.g. 10.0.0.0/24

Secondary IPv4 ranges

Add a secondary IPv4 range



Create Firewall Rules:

Google Cloud | student-00465

Network security

Cloud Armor
- DDoS dashboard
- Cloud Armor policies
- Adaptive protection
- Cloud Armor service tier

Cloud IDS
- IDS dashboard
- IDS endpoints
- IDS threats

Cloud NGFW
- Dashboard
- Firewall policies
- Threats
- Firewall endpoints

Secure access connect
- Realms Preview
- Attachments Preview

Cloud NSI
- Deployment groups
- Endpoint groups

Secure web proxy
- Web proxies
- SWP policies
- URL lists

Create a firewall rule

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. Learn more

Name *
allow-internal
Lowercase letters, numbers, hyphens allowed

Description
Allow internal traffic within VPC

Logs

Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more

- On
- Off

Network *
yousif-vpc

Priority *
1000 Compare
Priority can be 0–65535

Direction of traffic

- Ingress
- Egress

Action on match

- Allow
- Deny

Targets
All instances in the network

Source filter
IPv4 ranges

Source IPv4 ranges *
10.0.0.0/8 for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

Google Cloud | student-00465 | Firewall | Search

Open project picker (Ctrl O)

Network security

Cloud Armor
- DDoS dashboard
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Secure web proxy
- Web proxies
- SWP policies
- URL lists

DNS Armour
- Threat detection

Create a firewall rule

Name *
allow-http-https
Lowercase letters, numbers, hyphens allowed

Description
Allow HTTP and HTTPS traffic

Logs

Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more

- On
- Off

Network *
yousif-vpc

Priority *
1000 Compare
Priority can be 0–65535

Direction of traffic

- Ingress
- Egress

Action on match

- Allow
- Deny

Targets
Specified target tags

Target tags *
web-server

Source filter
IPv4 ranges

Source IPv4 ranges *
0.0.0.0/0 for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

Destination filter
None

Google Cloud | student-00465

Network security

Cloud Armor
- DDoS dashboard
- Cloud Armor policies
- Adaptive protection
- Cloud Armor service tier

Cloud IDS
- IDS dashboard
- IDS endpoints
- IDS threats

Cloud NGFW
- Dashboard
- Firewall policies
- Threats
- Firewall endpoints

Secure access connect
- Realms Preview
- Attachments Preview

Cloud NSI
- Deployment groups
- Endpoint groups

Secure web proxy
- Web proxies
- SWP policies
- URL lists

Create a firewall rule

Name *
allow-health-check
Lowercase letters, numbers, hyphens allowed

Description
Allow GCP Load Balancer health checks

Logs

Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more

- On
- Off

Network *
yousif-vpc

Priority *
1000 Compare
Priority can be 0–65535

Direction of traffic

- Ingress
- Egress

Action on match

- Allow
- Deny

Targets
Specified target tags

Target tags *
web-server

Source filter
IPv4 ranges

Source IPv4 ranges *
130.211.0.0/22 35.191.0.0/16 for example, 0.0.0.0/0, 1

Second source filter
None

Google Cloud | student-00465 | Firewall | Search

Network security

Cloud Armor
- DDoS dashboard
- Cloud Armor policies
- Adaptive protection
- Cloud Armor service tier

Cloud IDS
- IDS dashboard
- IDS endpoints
- IDS threats

Cloud NGFW
- Dashboard
- Firewall policies
- Threats
- Firewall endpoints

Secure access connect
- Realms Preview
- Attachments Preview

Cloud NSI
- Deployment groups
- Endpoint groups

Secure web proxy
- Web proxies
- SWP policies
- URL lists

DNS Armour
- Threat detection

Create a firewall rule

Name *
allow-ssh-iap
Lowercase letters, numbers, hyphens allowed

Description
Allow SSH through IAP only

Logs

Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more

- ○ On
- ◉ Off

Network *
yousif-vpc

Priority *
1000 Compare
Priority can be 0–65535

Direction of traffic

- ◉ Ingress
- ○ Egress

Action on match

- ◉ Allow
- ○ Deny

Targets
All instances in the network

Source filter
IPv4 ranges

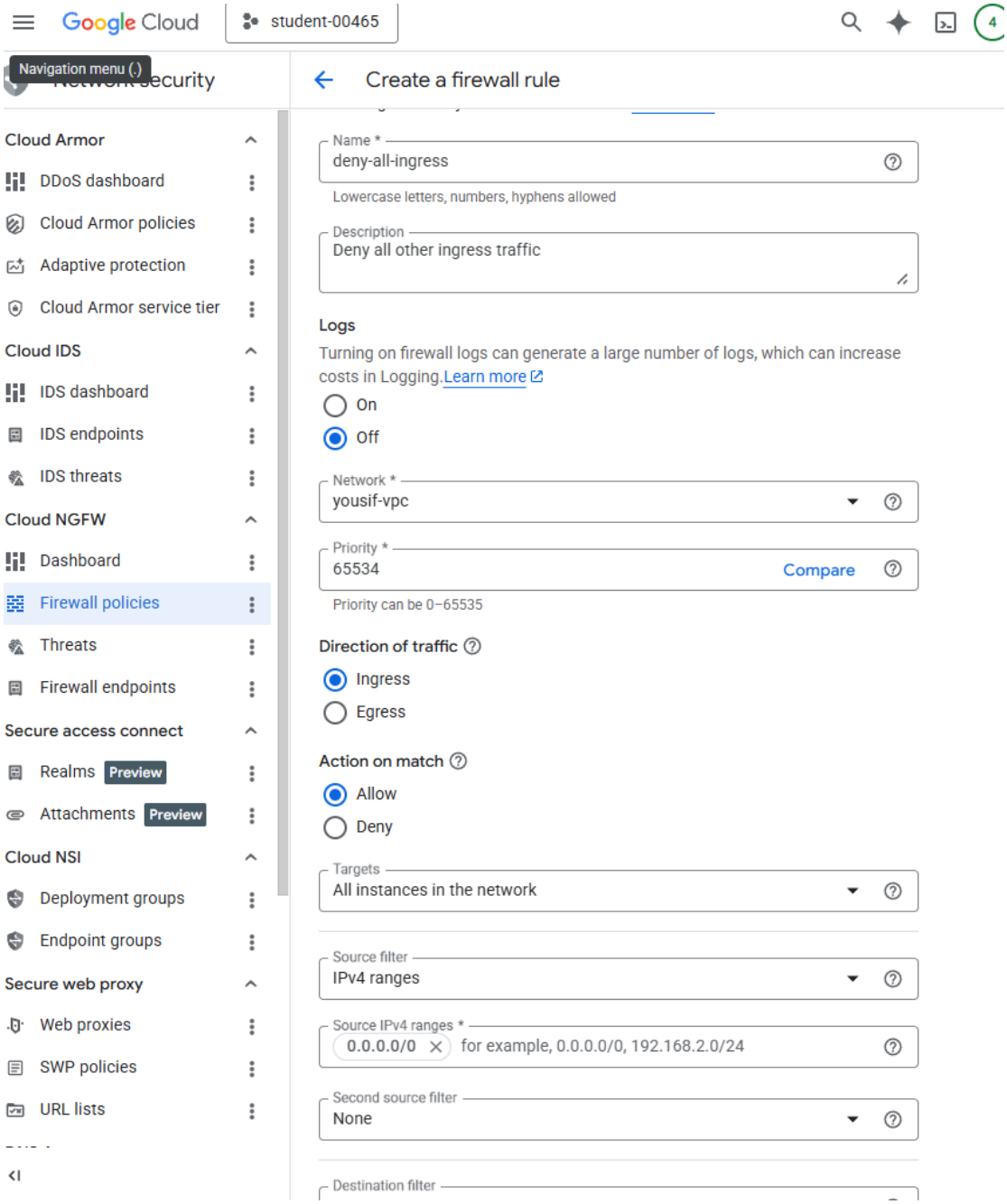
Source IPv4 ranges *
35.235.240.0/20 for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

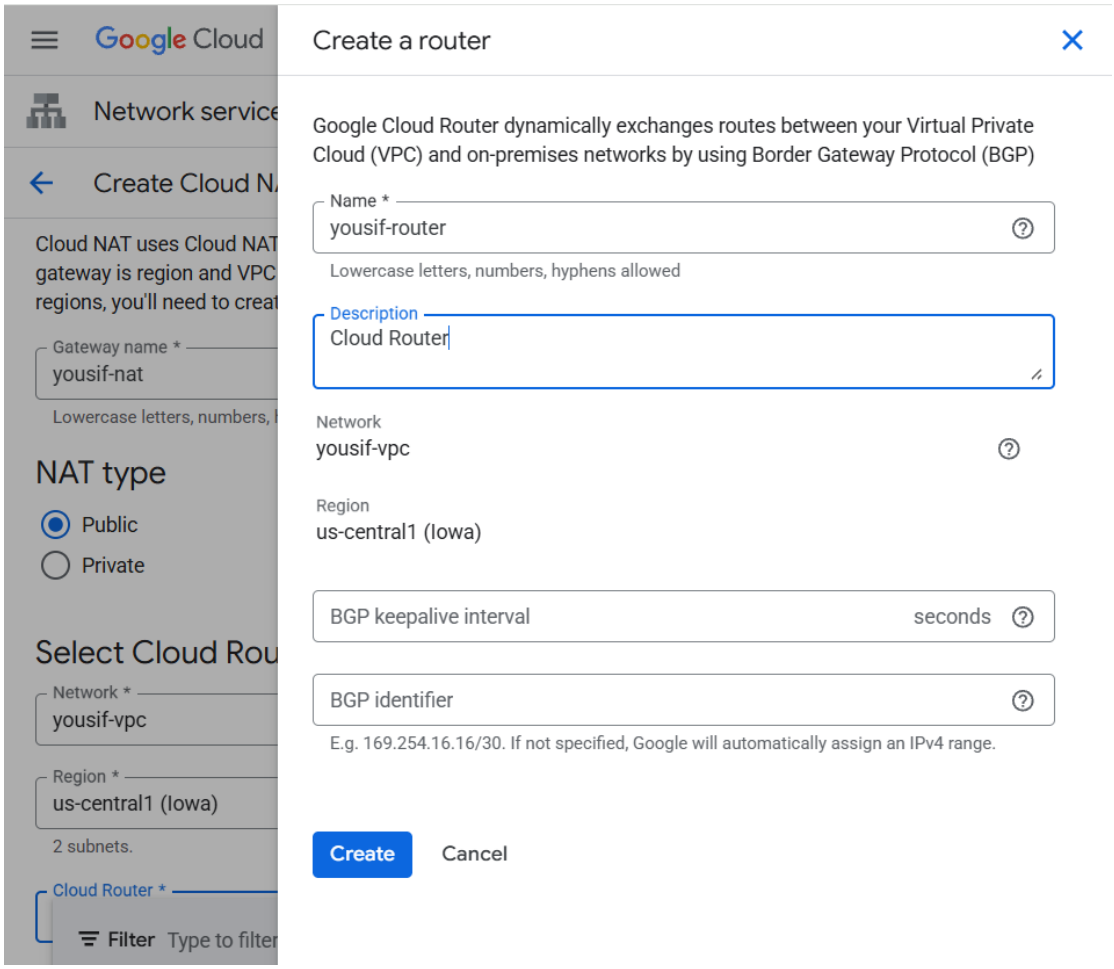
Destination filter
None

Protocols and ports

- ○ Allow all



Create Cloud NAT


Google Cloud
Network service
Create Cloud N
Cloud NAT uses Cloud NAT
gateway is region and VPC
regions, you'll need to creat
Gateway name *
yousif-nat
Lowercase letters, numbers,
NAT type
Public
Private
Select Cloud Rou
Network *
yousif-vpc
Region *
us-central1 (Iowa)
2 subnets.
Cloud Router *
Filter Type to filter
Create a router
Google Cloud Router dynamically exchanges routes between your Virtual Private Cloud (VPC) and on-premises networks by using Border Gateway Protocol (BGP)
Name *
yousif-router
Lowercase letters, numbers, hyphens allowed
Description
Cloud Router
Network
yousif-vpc
Region
us-central1 (Iowa)
BGP keepalive interval
seconds
BGP identifier
E.g. 169.254.16.16/30. If not specified, Google will automatically assign an IPv4 range.
Create
Cancel

Google Cloud | student-00465 | Cloud NAT

Network services

- Load balancing
- Cloud DNS
- Cloud CDN
- Cloud NAT
- Cloud Service Mesh (Traf...
- Service Directory
- Cloud Domains
- Private Service Connect
- Composite health
- SSL policies
- Service extensions

Marketplace

Release notes

Create Cloud NAT gateway

Gateway name *
yousif-nat
Lowercase letters, numbers, hyphens allowed

NAT type

- (●) Public
- ( ) Private

Select Cloud Router

Network *
yousif-vpc

Region *
us-central1 (Iowa)
2 subnets.

Cloud Router *
yousif-router

Cloud NAT mapping

Source endpoint type

- (●) VM instances, GKE nodes, serverless
- ( ) Managed proxy load balancers

Source IP version
IPv4 subnet ranges

IPv4 subnet ranges

Source subnets *
Custom

Select which subnets to map to the Cloud NAT gateway. Primary IP addresses are used by VM instances and secondary IP addresses are used by container pods.
Learn more

Subnets

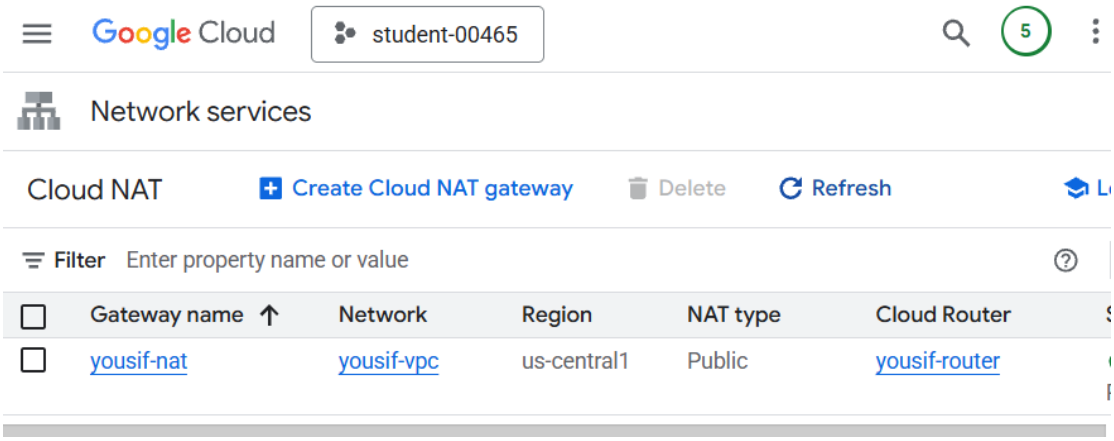
Subnet name 1: private-subnet (10.0.2...
IP ranges 1 *: Selected: All

+ Add subnet and IP range

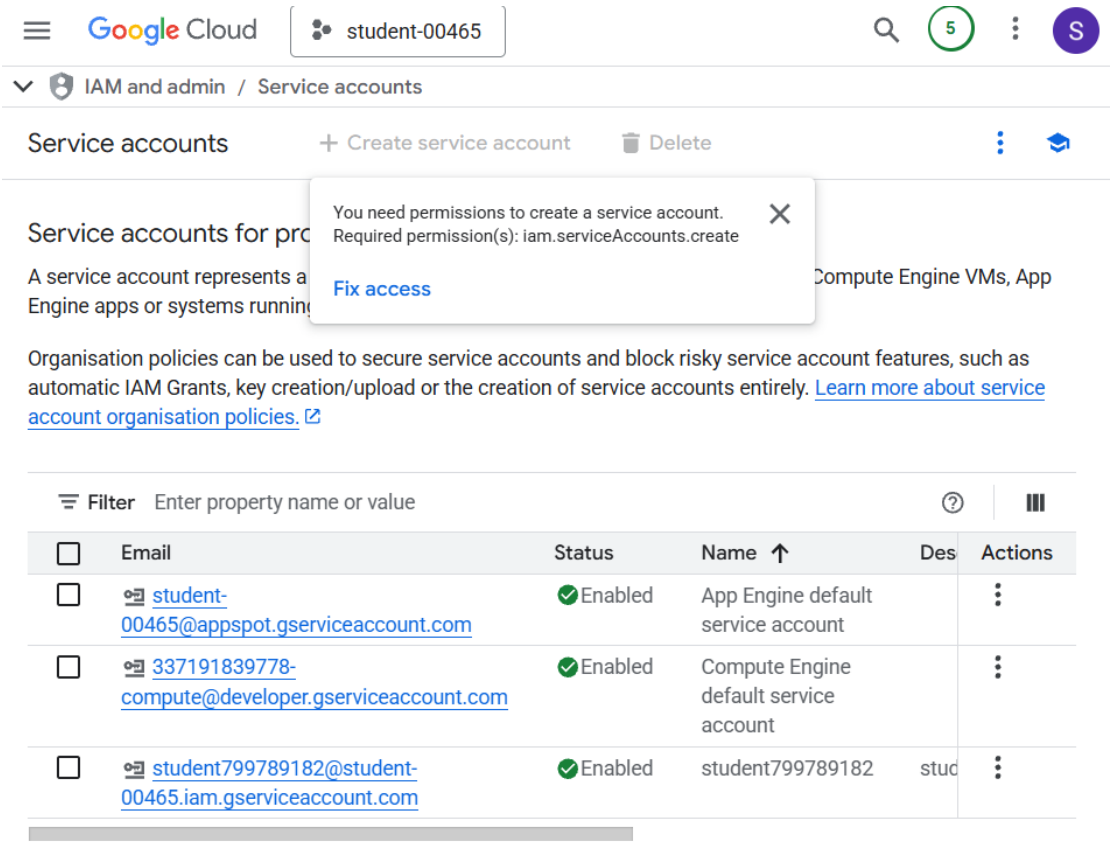
All VMs from selected subnet IP ranges will be able to communicate with Internet

Cloud NAT IP addresses
Automatic (recommended)

Network Service Tier



SECURITY SETUP



Create Instance Template

Compute Engine

- Overview
- Security risk overview

Virtual machines

- VM instances
- Instance templates
- Sole-tenant nodes
- Machine images
- TPUs
- Committed-use discou...
- Reservations
- Migrate to Virtual Mac...

Storage

- Disks
- Storage pools
- Snapshots
- Images
- Async replication
- Consistency groups

Instance groups

- Instance groups
- Health checks

VM Extension Manager

- VM extension policies

VM Manager

- Patch
- OS policies

Bare Metal Solution

- Servers

- Marketplace
- Release notes

Create an instance template

Describe a VM instance once and then use that template to create groups of identical instances Learn more

Name *
yousif-web-template

Manage tags and labels

Location

Global instance templates can be used in any region. To lower the impact of outages outside your region and gain data residency within your region, use a regional instance template. Learn more

- (●) Regional (recommended)
- ( ) Global

Region *
us-central1 (Iowa)

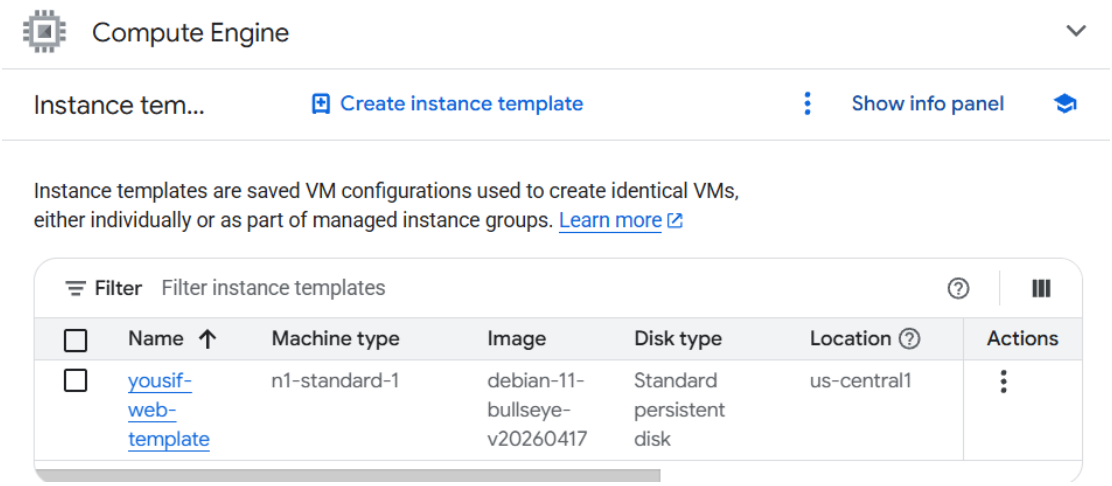
Generate machine type suggestions with Gemini New

Machine configuration

General purpose | Compute-optimised | Memory-optimised | Storage optimised | GPUs

Machine types for common workloads, optimised for cost and flexibility

	Series	Description	vCPUs
( )	C4	Consistently high performance	2 - 288
( )	C4A	Arm-based consistently high performance	1 - 96
( )	C4D	Consistently high performance	2 - 384
( )	N4	Flexible and cost-optimised	2 - 80
( )	N4A	Arm-based flexibility and cost optimisation	1 - 64
( )	N4D	Flexible and cost-optimised	2 - 96
( )	C3	Consistently high performance	4 - 192
( )	C3D	Consistently high performance	4 - 360
( )	E2	Low-cost, day-to-day computing	0.25 - 32
( )	N2	Balanced price and performance	2 - 128
( )	N2D	Balanced price and performance	2 - 224
( )	T2A	Scale-out workloads	1 - 48
( )	T2D	Scale-out workloads	1 - 60
(●)	N1	Balanced price and performance	0.25 - 96



Create Health Check

Compute Engine

Create a health check

Name *
http-health-check
Lowercase, no spaces.

Description
HTTP health check for web servers

Scope
Global
Regional

Protocol
HTTP

Port *
٨٠

Source regions
Used only for checking public endpoints attached to a Cloud DNS policy Learn more

PROXY protocol
None

Request path *
/

More

Logs
On

Create Cancel Equivalent command line Equivalent REST

Create Managed Instance Group

Compute Engine / Instance groups / Create instance group

Create Instance Group

Equivalent code

stateless serving and batch processing.

New managed instance group (stateful)
Automatically manage groups of VMs that have persistent data or configurations (such as databases or legacy applications).

New unmanaged instance group
Manually manage groups of load balancing VMs.

Name *
web-mig
Name is permanent

Description

Instance template *
yousif-web-template
n1-standard-1, debian-11-bullseye-v20260417, us-central1

Number of instances
Based on auto-scaling configuration

☐ Use resize request to create VMs all at once
Create and run VMs all at once when resources become available. Use as an alternative to VMs getting created on an individual basis. Learn more

Instance flexibility

Select multiple machine types for your instance group. This improves chances of cost-savings and obtaining resources. Learn more

Instance selections

Create groupings of multiple machine types and rank them in order of preference. Learn more

Add selections

Location

For higher availability, select multiple zones in a region instead of a single zone. Learn more

◉ Single zone
○ Multiple zones

Region *
us-central1 (Iowa)

Zone *
us-central1-a

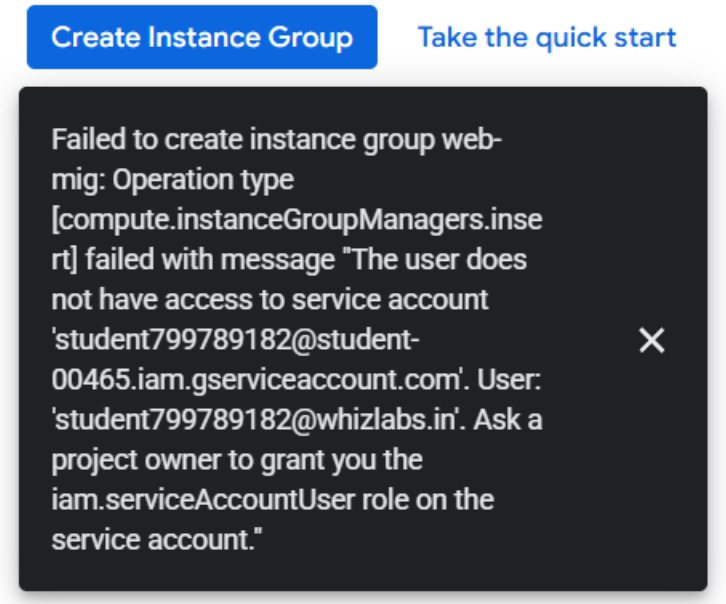
Auto-scaling

Use auto-scaling to automatically add and remove instances to the group for periods of high and low load. Learn more

Auto-scaling mode
On: add and remove instances to and from the group

Minimum number of instances *
2

Maximum number of instances *
5



LOAD BALANCER

Static External IP

VPC network / IP addresses / Reserve a static address

Reserve external static IP address

Name *
lb-ip
Lowercase letters, numbers, hyphens allowed

Description
Static IP for Load Balancer

Network Service Tier
- (•) Premium
  Current project-level tier, change
- ( ) Standard

IP version
- (•) IPv4
- ( ) IPv6

Type
- ( ) Regional
- (•) Global
  To be used with global forwarding rules. Learn more

[ ] Assign from IP collection

Static IP addresses not attached to an instance or load balancer are billed at a higher hourly rate. Pricing details

IP addresses Reserve external Reserve internal Show info panel

All | Internal IP addresses | External IP addresses | IPv4 addresses | IPv6 addresses

Filter Enter property name or value

Name	IP address	Access type	Region	Type	Version	Actions
lb-ip	34.110.145.50	External		Static	IPv4	

Create HTTP Load Balancer

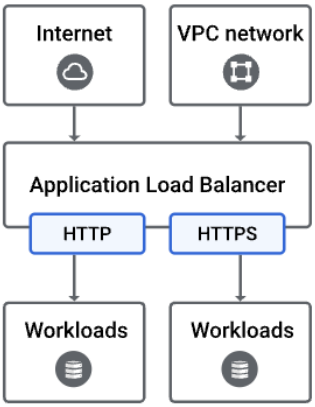
Network services

Create a load balancer

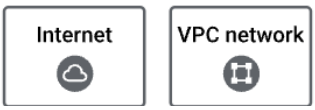
Learn

1 Type of load balancer

Application load balancer (HTTP/HTTPS)
Choose an application load balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic.



Network load balancer (TCP/UDP/SSL)
Choose a network load balancer when you need TLS offloading at scale, support for UDP, and exposing IP addresses to your applications.



- Load balancing overview
- Application load balancer
- Proxy network load balancer
- Passthrough network load balancer
- Public facing vs internal
- Global vs regional
- Proxy vs passthrough
- GCP load balancer pricing

Create global external application load balancer

Load balancer name *

Lowercase, no spaces.

- Frontend configuration
- Backend configuration
- Routing rules
- Review and finalise (optional)

New Frontend IP and port

Name

frontend

Lowercase, no spaces.

Description

Protocol

HTTP

Select HTTPS to support clients that support HTTP/2. The load balancer automatically offers HTTP/2 as part of the TLS handshake.

Network Service Tier

Premium

Global HTTP(S) load balancing only supports the Premium Network Service Tier. Learn more

IP version

IPv4

IP address

lb-ip

Port *

80

Google Cloud

Network service

Create global ex

Load balancer name *

Lowercase, no spaces.

Frontend configura

Backend configura

Routing rules

Review and finalise

Create Cancel

Create back-end service

Name *
web-backend
Lowercase, no spaces.

Description

Backend type
Instance group

Protocol
HTTP

Named port *
http

Timeout *
٣٠ seconds

IP address selection policy
Only IPv4

Health check *

Backends

Add a backend

Cloud CDN

Enable Cloud CDN

Create Cancel

Create global external application load balancer

Load balancer name *

Lowercase, no spaces.

- Frontend configuration
- Backend configuration
- Routing rules
- Review and finalise (optional)

Review and finalise

Frontend

Protocol	IP:Port	Certificate	Certificate map
HTTP	34.110.145.50:80	-	

Routing rules

Hosts	Paths	Backend
All unmatched (default)	All unmatched (default)	web-backend

Backend

Backend services

1. web-backend

Endpoint protocol	HTTP
Timeout	30 seconds
IP address selection policy	Only IPv4
Health check	httphealthcheck
Cloud CDN	Disabled
Logging	Enabled (sample rate: 1) All optional fields excluded
Backend security policy	None

Show advanced

Backends

This backend service has no backends configured

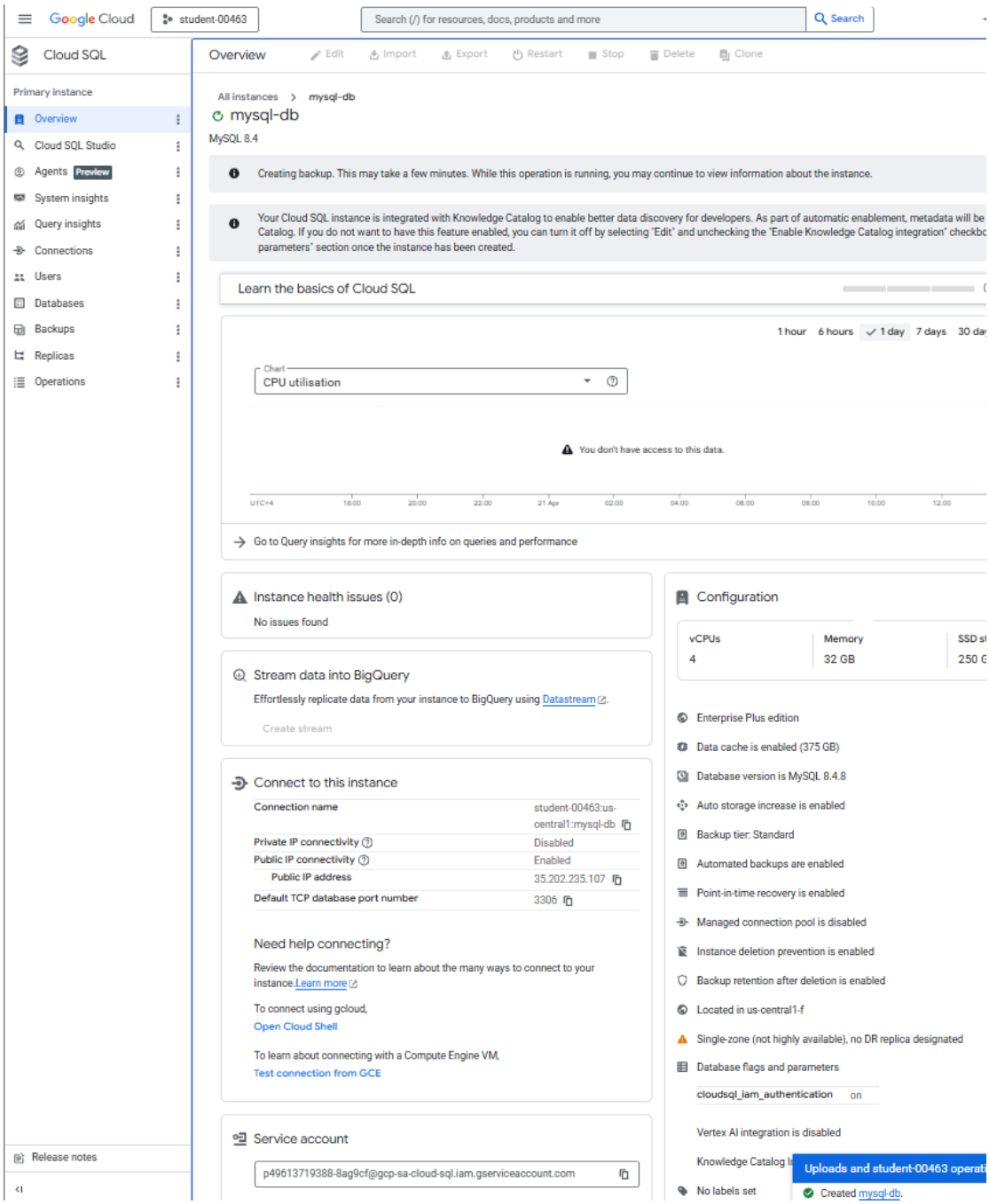
Load balancers | Backends | Frontends | Service LB policies

Filter Enter property name or value

Name	Load balancer type	Access type	Protocols	Region	Actions
yousif-load-balancer	Application	External	HTTP		

DATABASE LAYER

Create Cloud SQL Instance



Google Cloud

Cloud SQL

Primary instance

- Overview
- Cloud SQL Studio
- Agents Preview
- System insights
- Query insights
- Connections
- Users
- Databases
- Backups
- Replicas
- Operations

Release notes

Add a user account to instance mysql-db

Choose how to authenticate

You can manage access to this instance using Cloud IAM or MySQL built-in authentication. Learn more

(●) Built-in authentication

Creates a new username and password specific to this instance. User account will have **cloudsqlsuperuser** root access, but you can customise that later as needed. Learn more

Username *

admin

Password (optional)

••••••••••••••••

Generate

- Must contain an uppercase letter, lowercase letter, number and non-alphanumeric character
- Must not contain the username
- Must be at least 8 characters long

Password policy

Host name

(●) Allow any host (%)

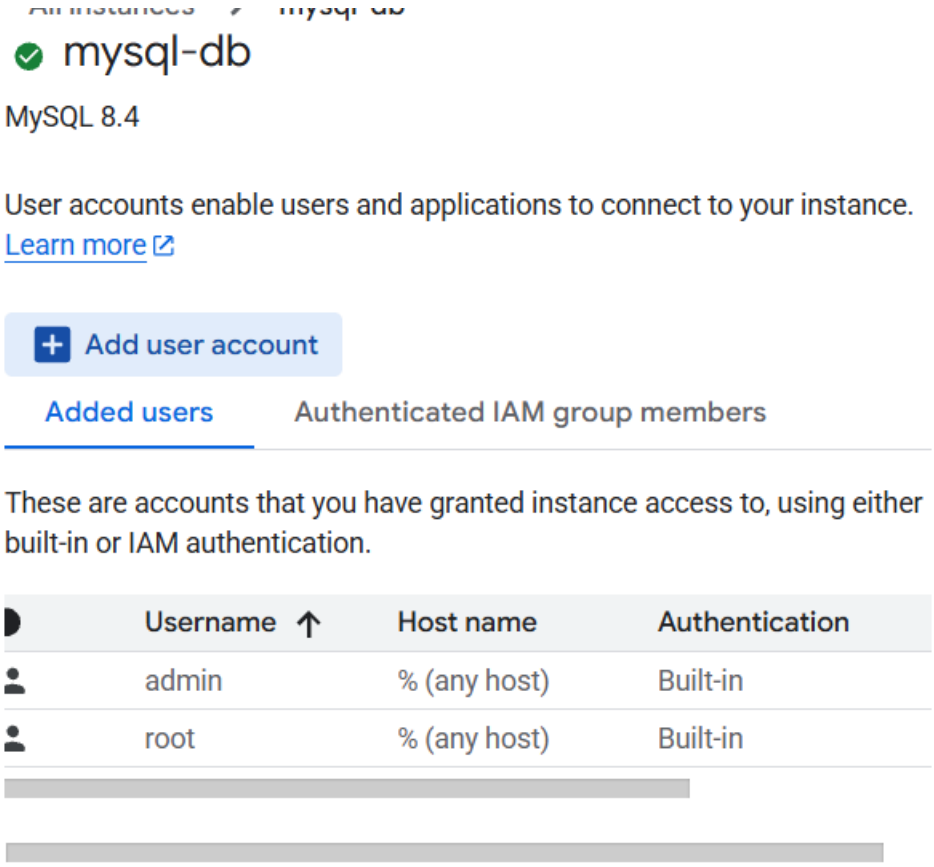
( ) Restrict host by IP address or address range

Users created with built-in authentication have the same privileges as the **root** user. Learn more

( ) Cloud IAM

Adds an existing IAM principal as a user account on this instance. The IAM principal can be an IAM user, service account or a group. The IAM principal must have cloudsql.instances.login permission using the predefined Cloud SQL Instance User IAM role or custom IAM role in the IAM policy of the instance's project.

Add Cancel



MONITORING & LOGGING

Monitoring

Notification channels

Google Chat Preview — Add New

No Google Chat channels configured

PagerDuty Services — Add New

No PagerDuty services configured

PagerDuty Sync Beta — Add New

No PagerDuty Sync channels configured

Slack — Add New

No Slack channels configured

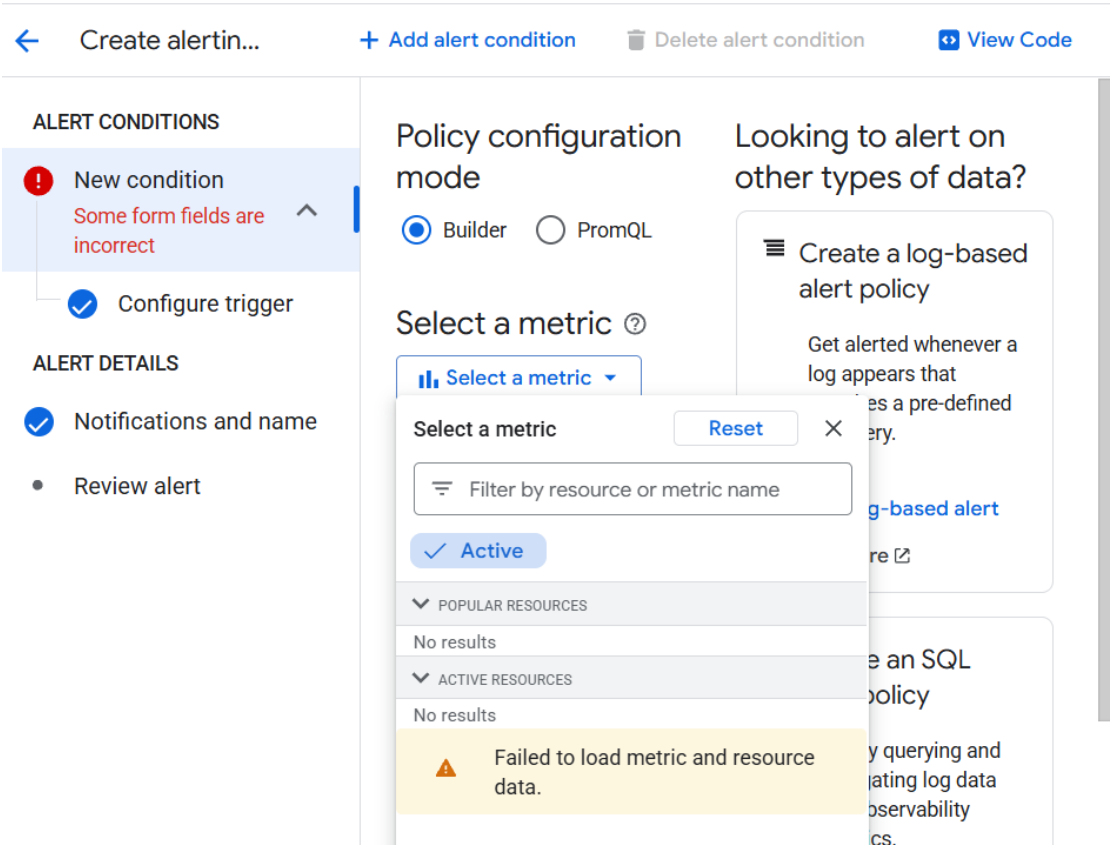
Webhooks — Add New

No webhook channels configured

Email — Add New

Filter Filter email addresses

Email	Display Name
yousif2alrisi@gmail.com	Alert Email


Create alertin...
Add alert condition
Delete alert condition
View Code
ALERT CONDITIONS
New condition
Some form fields are incorrect
Configure trigger
ALERT DETAILS
Notifications and name
Review alert
Policy configuration mode
Builder
PromQL
Select a metric
Select a metric
Select a metric
Reset
Filter by resource or metric name
Active
POPULAR RESOURCES
No results
ACTIVE RESOURCES
No results
Failed to load metric and resource data.
Looking to alert on other types of data?
Create a log-based alert policy
Get alerted whenever a log appears that
es a pre-defined
ery.
g-based alert
re
e an SQL
olicy
y querying and
ating log data
bservability
cs.

Configure wid...

Last 1 hour GST

Apply Cancel

Queries Add query Create ratio Auto-run

A Filters CPU Utilization PromQL

This monitoring filter cannot be edited in standard mode because it could not be parsed.

Grouping None by None Alignment function Mean

Results Chart Table Both

Query A: Permission denied (or the resource may not exist).

UTC+4 14:20 14:30 14:40 14:50 15:00

0 time series

Display

Widget type

Line chart

Widget title

CPU Utilization [MEAN]

Analysis mode

Analysis mode

Standard mode

Compare to past

Threshold line

Y-axis assignment

Data	Unit	Y-axis
A	unitless	Right

Y-axis labels

Right Y-axis label